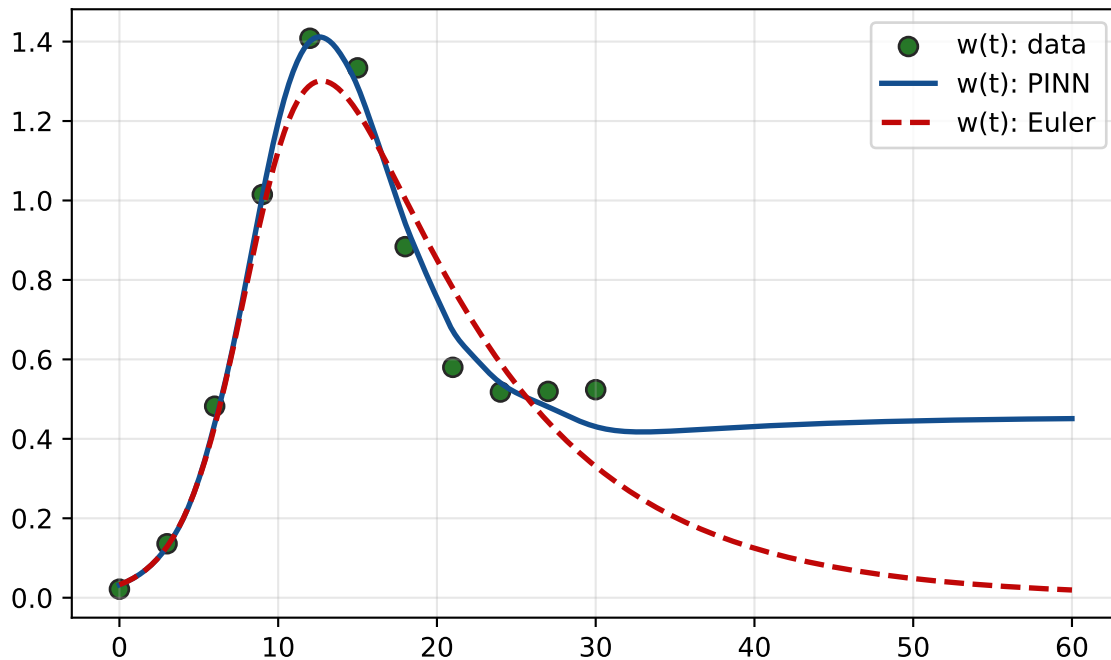
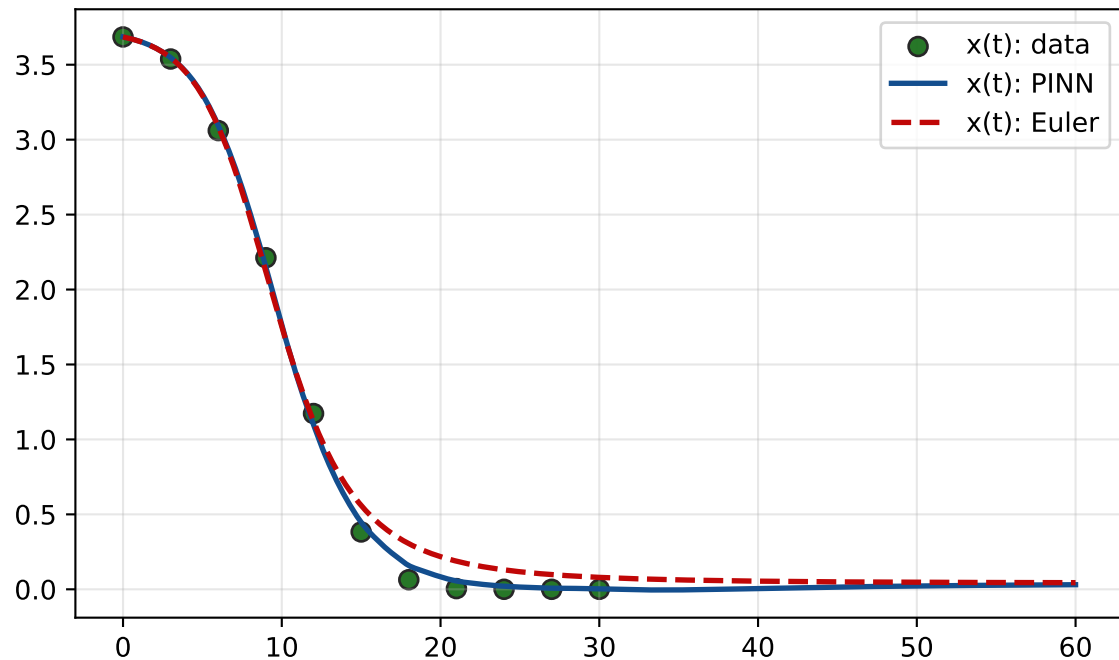


# E3 | PINN vs Euler

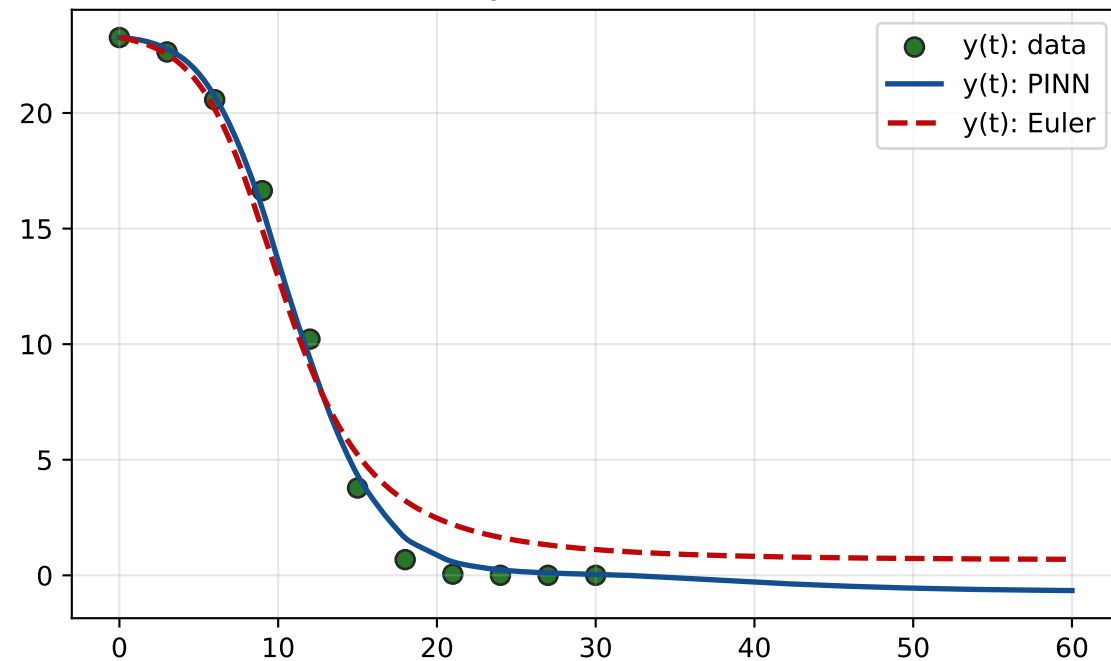
## w(t): Biomass



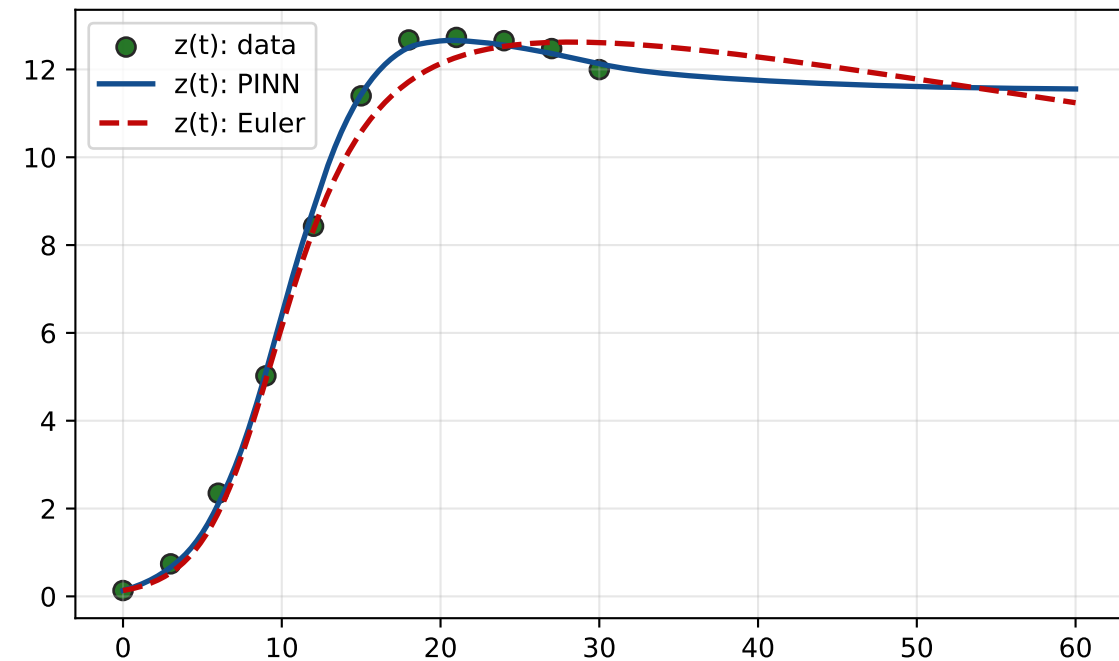
## x(t): Glucose



## y(t): Fructose



## z(t): Ethanol



E3

R2=0.9866, R2\_adj=0.9840, RSS=2.595e+01, AIC=-6.120e+00

cond(JTJ)=7.69e+04, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 1.768903e-02 | 2.390209e-03 | 4.843024e-03 | 1.284601e-02  | 2.253205e-02 | 8.348110e-09 |
| p2    | 9.957378e-02 | 1.032772e-01 | 2.092595e-01 | -1.096857e-01 | 3.088333e-01 | 3.412347e-01 |
| p3    | 7.373611e-03 | 5.021703e-03 | 1.017494e-02 | -2.801325e-03 | 1.754855e-02 | 1.504604e-01 |
| p4    | 1.812900e-01 | 8.345958e-02 | 1.691052e-01 | 1.218478e-02  | 3.503951e-01 | 3.632242e-02 |
| p6    | 1.431410e-01 | 4.378632e-02 | 8.871951e-02 | 5.442151e-02  | 2.318605e-01 | 2.335736e-03 |
| p8    | 7.033238e-02 | 2.229754e-02 | 4.517910e-02 | 2.515327e-02  | 1.155115e-01 | 3.189546e-03 |
| p9    | 5.031424e-03 | 6.612617e-03 | 1.339843e-02 | -8.367010e-03 | 1.842986e-02 | 4.515494e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 3.262115e-02 |
| x0 (fixed)     | 3.685297e+00 |
| y0 (fixed)     | 2.326363e+01 |
| z0 (fixed)     | 1.326396e-01 |
| W0 (fixed)     | 0.000000e+00 |